Software Process Management

Lecture 6

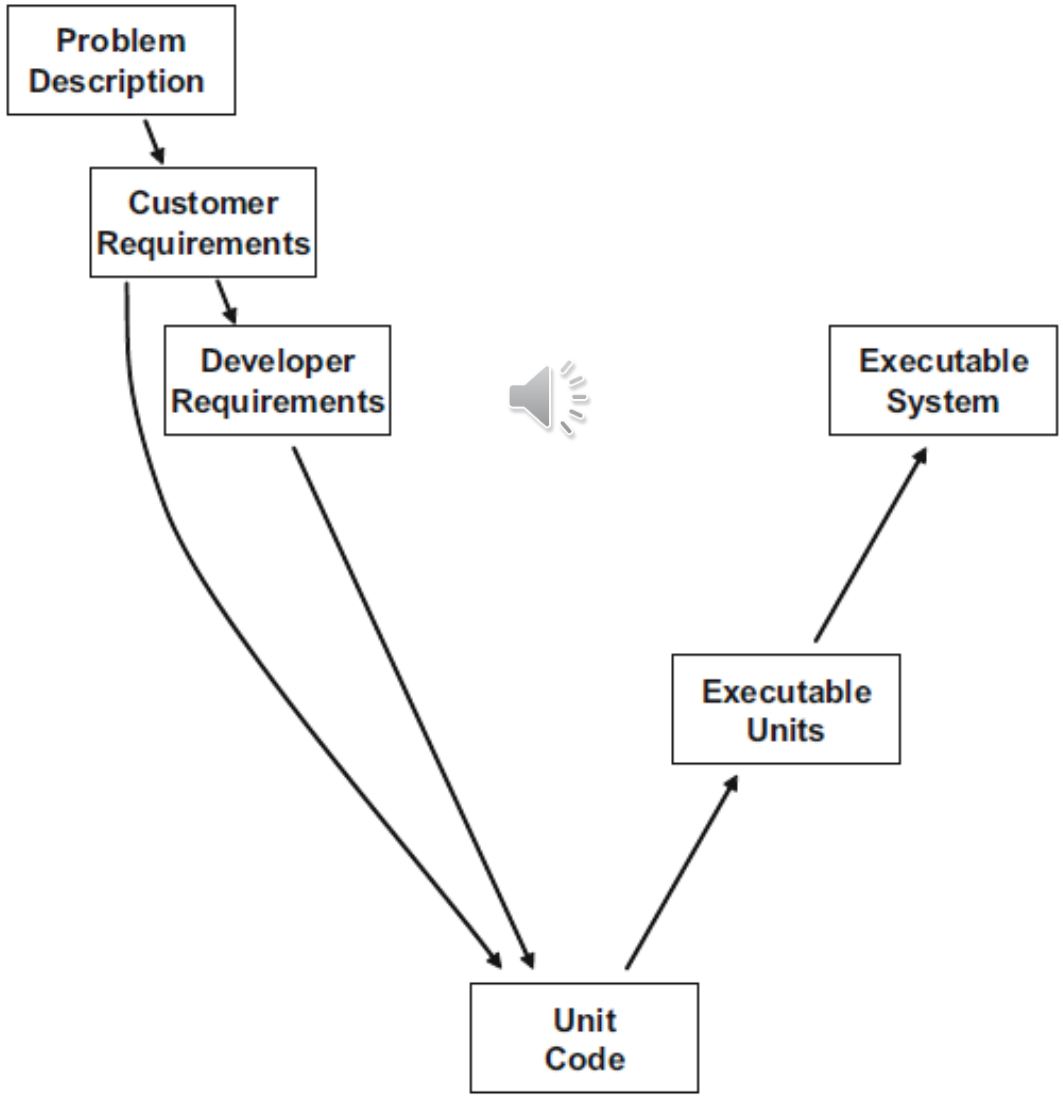
What have we learnt so far?

- Prescriptive Process Models
- Product Process Relationship
- Prerequisites
- Prescriptive Process Model Classes
  - Continue..

Prototyping Model

- Prototyping concentrates on the development of an executable version of the system that fulfils a limited number of requirements.
- The primary goal of building prototypes is to gain initial experience (e.g., regarding unclear requirements or difficult design aspects).
- Here we focus on Throwaway Prototyping

Prototyping Model



- Requirements typically directly lead to code, skipping all the other steps encountered, for example, in the waterfall or the iterative enhancement model.

- Prerequisites to use Prototyping Model
- One must possess a high level of experience with the development techniques used, because all common assistance in the form of designs, modules, etc., is usually missing.
- The technical infrastructure for creating and evaluating a prototype must be available

Advantages of Prototyping Model

- A prototype can be developed when the final properties are not entirely clear.
- The direct contact between customer and developer reduces misunderstandings.
- Inconsistent requirements are discovered earlier, either by the developer or by the customer who evaluates the prototype.

Advantages

- Comprehensive version and configuration management is not necessary because the code produced is not meant to exist for a long time.
- The risk of project failure (e.g., developing a product that does not satisfy the customer requirements) is reduced due to the early involvement of the customer.
- In some cases, prototypes may even be used for evaluating business models, before a lot of money is spent on developing the real software.

Challenges

- There is an inherent risk that side effects are not sufficiently considered, especially non-functional requirements such as reliability or safety.
- The customer (or the developer) considers the prototype as the first version of the system, and that the system will be evolved from the prototype, potentially leading to poorly documented, badly architected systems.
- The prototyping phase of a project may also induce higher costs compared to a non-prototyped approach.

The Spiral Model

- The spiral model was first published in 1986 by Barry Boehm
- Represents a risk-driven approach, i.e., the assessment of risks determines the next project phase.
- Combines aspects of the waterfall model, the iterative enhancement model, and prototyping.

- Prerequisites for applying the spiral model is the capability to identify and assess risks, as well as a project context that allows for changing the process model during project run-time.

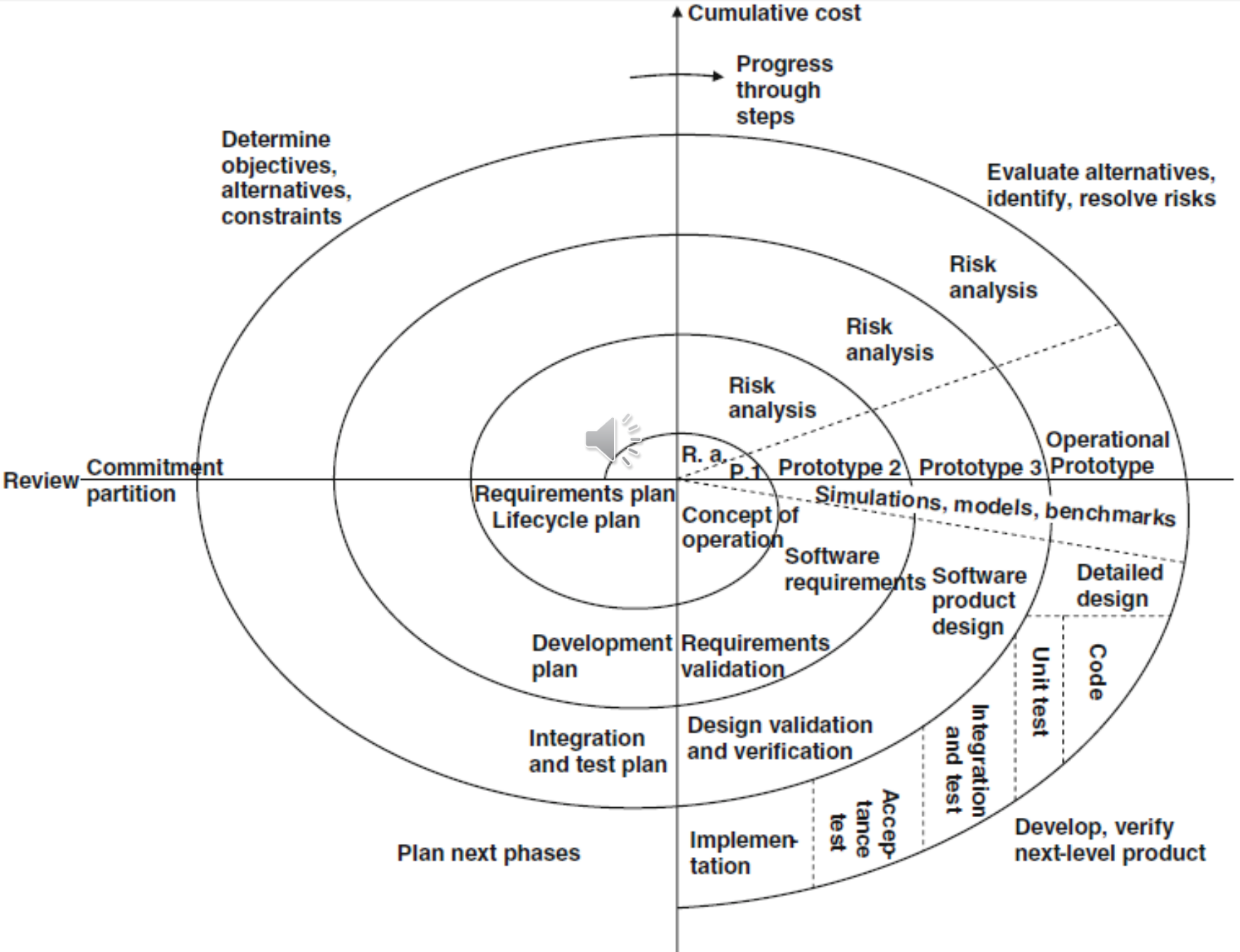
The Spiral Model

- The first step of each spiral cycle identifies the objectives of the product part being elaborated (e.g., performance or functionality), the different alternatives for implementing the product part (e.g., different designs or reuse of existing components), and the constraints for each of the identified alternatives (e.g., cost or schedule).

- The next step evaluates the identified alternatives and identifies and resolves risks that come with the different alternatives.

- The third step is determined by the risks that remain after the second step.
- During the third step, the development approach that is suited best for the risks is chosen.
- This may be the construction of a prototype or the application of a strict waterfall process for implementing a clearly defined feature.

- Finally, the next phases are planned, and the complete cycle is reviewed by the stakeholders.


Cumulative cost
Progress through steps
Determine objectives, alternatives, constraints
Evaluate alternatives, identify, resolve risks
Risk analysis
Risk analysis
Risk analysis
Review
Commitment partition
R. a.
P.1
Prototype 2
Prototype 3
Operational Prototype
Requirements plan
Lifecycle plan
Concept of operation
Simulations, models, benchmarks
Software requirements
Software product design
Detailed design
Development plan
Requirements validation
Unit test
Code
Integration and test
Integration and test plan
Design validation and verification
Accep-tance test
Implemen-tation
Plan next phases
Develop, verify next-level product

Advantages

- The third step accommodates features of other lifecycle process models as needed.
- The spiral model is therefore very flexible.
- The explicit consideration of risks avoids many of the difficulties of other process models.
- Similarly, unattractive alternatives are identified and eliminated early.

- The spiral model forms a single approach for software development and maintenance, whereas other models often concentrate on one or the other.

Challenges

- For contract software, the spiral model is difficult to apply because individual project phases are not fully determined during project setup.

- Spiral model relies heavily on the organization's expertise with respect to risk assessment—
- Therefore, a bad risk assessment may lead to the selection of bad alternatives or development approaches.

Important

- Boehm's spiral model embraces risk as its central aspect, and chooses the best approach for every iteration.

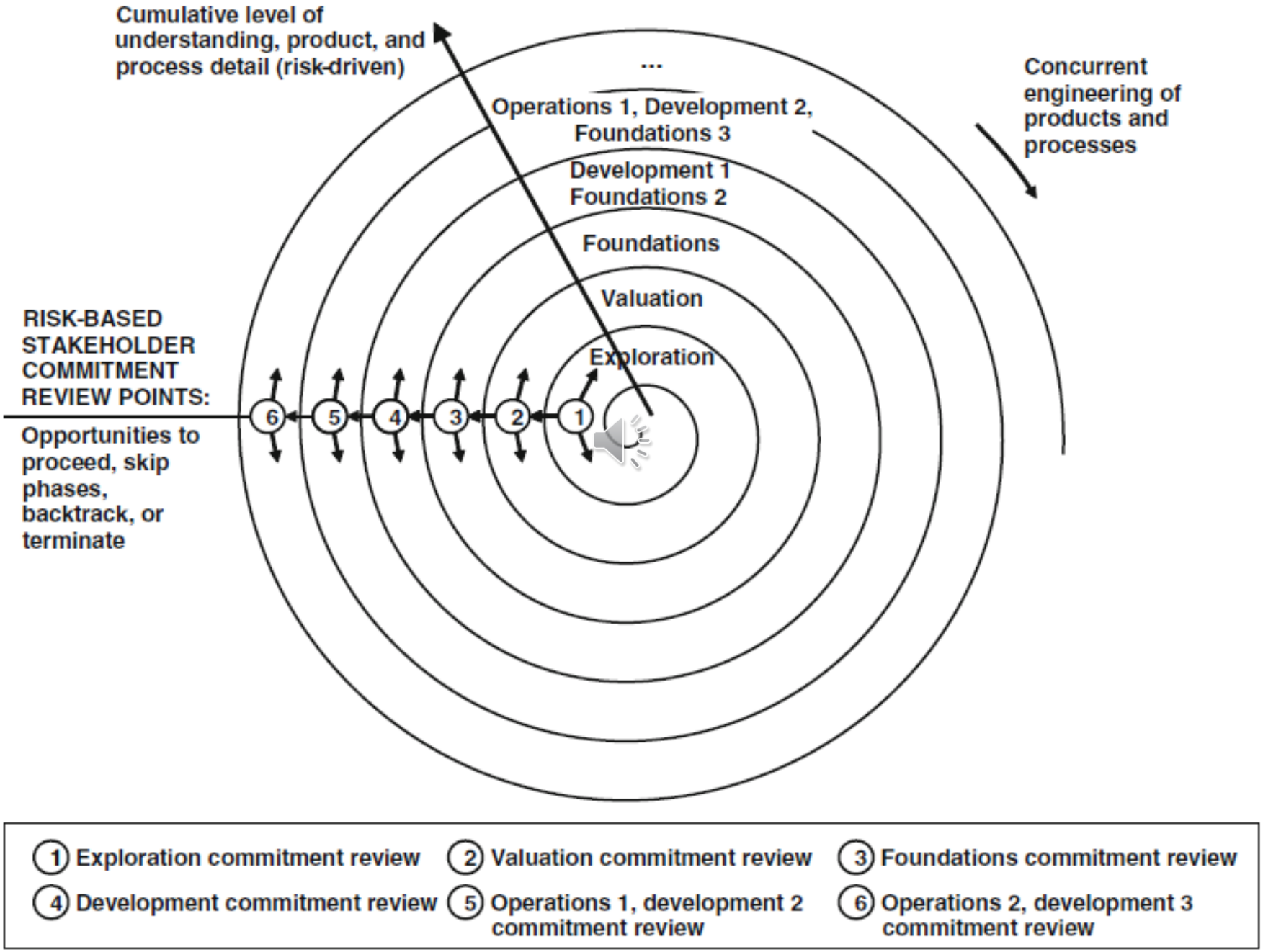
- This distinguishes it from other lifecycle models, which consider risks, but do not change the basic model accordingly.

The Incremental Commitment Spiral Model

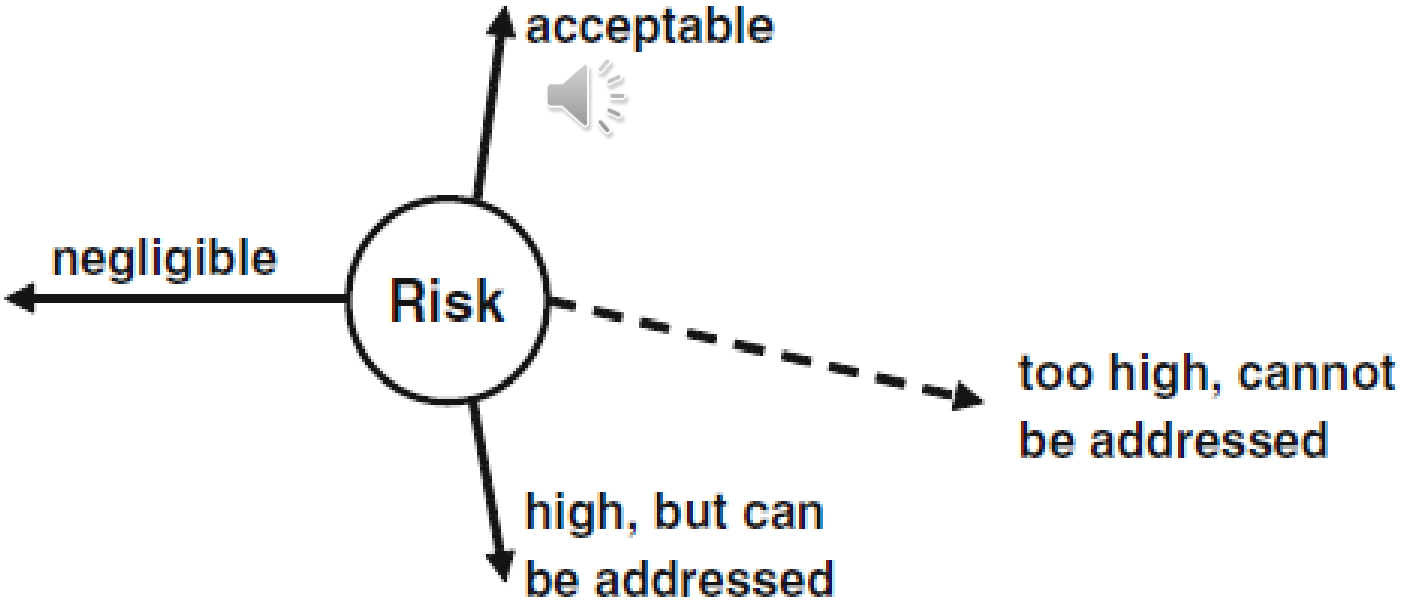
- The original spiral model demands that for every spiral, a project should identify its objectives, constraints, and alternative solution approaches;
- Evaluate these alternatives; and then
- Decide on the next steps based on the identified risks.

- However, this has often been misunderstood;
- The choice between the different development process alternatives has often been neglected, leading to “unrolling” of the spiral model into a strict waterfall model or perceiving it as a simple incremental development process without any decision making in between.

- To address these issues a modified version of the model was developed – ICSM (incremental commitment spiral model)
- Similar to the original spiral model, it consists of a spiral covering the development lifecycle of a system.
- However, it makes the risk-driven decisions explicit, emphasised by the stakeholder commitment reviews 1–6.


Cumulative level of understanding, product, and process detail (risk-driven)
...
Operations 1, Development 2, Foundations 3
Development 1 Foundations 2
Foundations
Valuation
Exploration
Concurrent engineering of products and processes
RISK-BASED STAKEHOLDER COMMITMENT REVIEW POINTS:
Opportunities to proceed, skip phases, backtrack, or terminate
6
5
4
3
2
1
1 Exploration commitment review
2 Valuation commitment review
3 Foundations commitment review
4 Development commitment review
5 Operations 1, development 2 commitment review
6 Operations 2, development 3 commitment review

- Each of these reviews constitutes a decision point with four possible exits.



Risk-based decisions

Possible Exits

- If the identified risks are acceptable for the stakeholders and well covered by appropriate mitigation plans, the project will continue into the next spiral.
- An example would be a software system that does not reach its intended throughput, but that scales well with the number of CPUs—the mitigation plan would be to add additional CPUs in order to reach the target throughput.

- If the identified risks are high, but expected to be addressable, the project will remain in the current spiral until the risks are resolved or covered by appropriate mitigation plans.
- An example would be working out safety cases for a safety-critical system or producing acceptable versions of missing risk mitigation plans.

Possible Exits

- If the identified risks are negligible, the project can skip the next spiral(s). This means, for example, that if the risks are determined to be negligible during the Exploration spiral, both the Valuation and Foundations spirals can be skipped, and the project can go directly to the Development spiral.
- For example, this can happen if the Exploration spiral finds that the solution can be easily produced by tailoring a COTS package rather than developing a custom system.

Possible Exits

- If the identified risks are determined to be too high or cannot be addressed, the project should be terminated or re-scoped (back exit).
- An example would be that the market window for the product under development has already closed—hence, the project should either be terminated, or it should be re-scoped to address a different market sector whose market window is still open.

- Each spiral concurrently addresses all of the activities of product development:
- Requirements (objectives and constraints) and solutions (alternatives);
- Products and processes; hardware, software, and human factors aspects; and
- Business case analysis of alternative product configurations or product line investments

are all being considered in every spiral.